

Processing THz Signals with GHz Electronics

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Abstract. We present the fundamentals and latest findings on the generation, processing and reception of THz-bandwidth signals using low-bandwidth electronics and orthogonal base functions for communication and the acceleration of photonic signal processing. ©2026 The Author(s)

Introduction

To cope pace with bandwidth hungry applications and services like artificial intelligence, gaming and streaming, new ways to generate, transmit and process high data rate signals are required. Especially the power consumption and restricted bandwidth of typical complementary metal oxide semiconductor (CMOS) devices presents a severe bottleneck [1].

A very promising approach is to combine the high-bandwidth signal from parallel low-bandwidth sub-signals in the transmitter before transmission, and to divide it into these sub-signals in the receiver. The processing of these sub-signals can then be carried out with parallel, integrated low-bandwidth CMOS electronics. This may offer a higher signal-to-noise and distortion (SINAD) ratio and therefore a higher effective number of bits (ENOB) [2]. Thus, it may enable the usage of modulation formats with a higher spectral efficiency for the link.

Due to these advantages, many different methods for bandwidth down-conversion, like time-interleaving by sub-sampling with a mode-locked laser (MLL) [3], time-stretching [4, 5], and spectrum stitching, or slicing [6, 7] have been shown. Some of these methods can be combined with optical frequency combs to achieve signals with bandwidths in the THz domain [7 – 9]. However, sub-sampling requires a precisely stabilized MLL, which can be quite big and complex. For time stretching the required second order dispersion is typically accompanied by higher orders, which leads to a distortion of the signal. And most of the frequency slicing or stitching methods lead to a complex post- or pre-processing.

Here we review methods which are based on orthogonal base functions, which ensure the inter-symbol-interference (ISI) free combination of and division into the sub-signals.

Orthogonal Base Functions

The basic idea of using orthogonal base functions (OBF) for the combination of a high bandwidth signal from sub-signals (sub1-N) at the

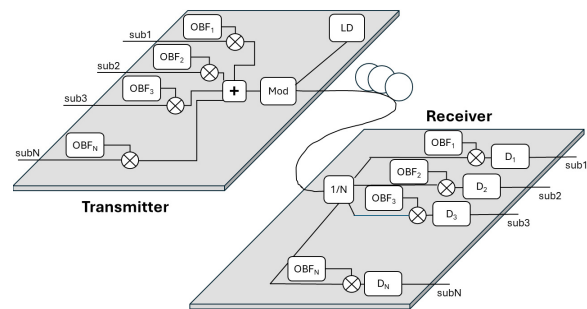


Fig. 1: Orthogonal signal processing. LD; laser diode, Mod; modulation, D; detector, sub; sub-signal, $1/N$; power splitting, OBF; orthogonal base function.

transmitter and the division into these base functions at the receiver after transmission is shown in Fig.1. In the transmitter the low-bandwidth, low-sampling-rate sub-signals are multiplied with the respective OBF. As sub-signal any waveform generated by a low-bandwidth digital to analog converter (DAC), or a direct analog input is possible. Due to orthogonality, the weighted sub-signals can be simply added up to build the high-bandwidth signal, which is transmitted. In the receiver, after power-splitting, the high-bandwidth signal is multiplied with the respective OBF and the required integration for the demultiplexing of the orthogonal signals is carried out in a detector with the bandwidth of the sub-signal. If the detector bandwidth is higher, a software or electrical filter in the baseband is required. For N orthogonal base functions, each sub-signal in the transmitter and receiver has $1/N^{\text{th}}$ of the bandwidth and sampling rate, which drastically reduces the requirements on the electronics. The combination and division can be done optically or in the electric domain. In the optical domain a modulator and in the electrical a mixer can be used for the multiplication. As an example, Fig. 1 shows the electrical combination of the sub-signals in the transmitter [10], and the optical division in the transceiver [11]. The processing in Fig.1 can as well be carried out by software or electronics but, without bandwidth improvement.

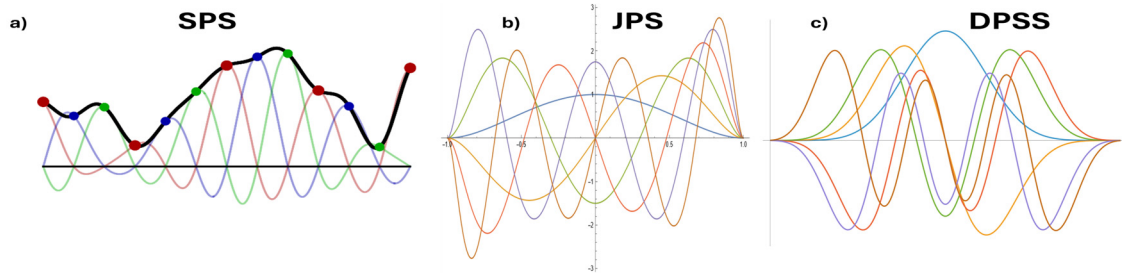


Fig.2: Orthogonal base functions: (a) three weighted sinc pulse sequences (red, green and blue), (b) first six Jacobi pulse sequences (c) first six discrete prolate spheroidal, or Slepian sequences.

As base function in principle any orthogonal function can be used. The arguably most important orthogonal function is that of a sinc pulse. According to the sampling theorem, any bandwidth limited signal can be seen as the unlimited superposition of sinc pulses, where each pulse is weighted with a single sampling point of the signal. To fulfil the orthogonality condition, the next sinc pulse is time-shifted to the zero-crossing of the previous.

Sinc pulses, however, are unlimited in time, and therefore not practicable. Sinc pulse sequences (SPS), as shown in Fig.2 (a), are a superposition of single sinc pulses, with the next pulse in the N -th zero crossing of the previous [12]. In complete contrast to single sinc pulses, SPS are a rectangular frequency comb in the spectral domain, which can be generated quite easily. The simplest SPS consists of a direct current (DC) and a sinusoidal. Around the carrier this corresponds to a three-line comb (two sidebands plus carrier, $N = 3$). Therefore, such a three-line comb only requires a DC shifted sinusoidal [8]. Figure 2 (a) shows the construction of a signal (black curve) from three, three-line SPS (red, blue and green). In contrast to single sinc pulses, which are multiplied with the single sampling points, SPS are multiplied with the analog sub-signals. Each single SPS carries $1/3^{\text{rd}}$ of the sampling points, and reduces the bandwidth and sampling rate requirements of the electronics by 3.

In Fig.2 (b) the first six orders of Jacobi pulse sequences (JPS) are shown [13]. In contrast to SPS which need a time-shift to be orthogonal, JPS are orthogonal in the same time interval. Therefore, with N JPS N complex values can be multiplexed into the same time interval [13]. Such a time-multiplexing can as well be achieved with Hermite-Gaussian polynomial sequences (HGS). However, for HGS the required time interval and the bandwidth increases with the number of superimposed sub-signals [14, 15]. For JPS instead, the time interval is independent of the number, but the bandwidth of the multiplexed signal increases. However, this bandwidth increase

is not much higher than the Nyquist bandwidth, which would be required for the aggregated signal [13]. Any number of JPS orders can be generated quite simply by a recursive method with analog electronics like multipliers and adders [13]. In contrast to a digital processing, analog electronics is not dependent on sampling rates and offers bandwidth in excess of 100 GHz [16]. For the electronics in the transmitter and receiver, however, only the low-bandwidth of the sub-signals is needed.

An optimum confinement of the multiplexed signal in the time- and frequency domain is enabled by discrete prolate spheroidal, or Slepian sequences (DPSS). The first six orders of DPSS are shown in Fig.2 (c). Like JPS, DPSS are perfectly confined in time and can be used to multiplex N complex values into the same time frame. But their special advantage is that they are optimally confined in the spectral domain too [17]. However, since JPS were already well confined in the spectral domain, the possible bandwidth improvement with DPSS is not much. Additionally, compared to JPS the generation of DPSS requires quite complex digital electronics.

Results

For communications especially SPS offer many unique advantages. Figure 3 shows how SPS can be used to generate high-bandwidth signals from low-bandwidth sub-signals in an integrated modulator [18].

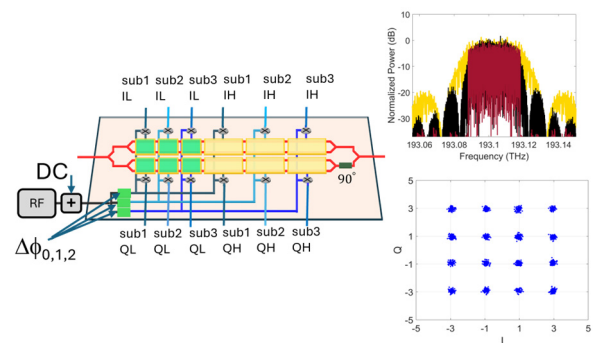


Fig.3: Segmented, two stage (L and H) IQ modulator for the generation of high-bandwidth signals with SPS.

The main idea behind this integrated modulator is to simultaneously reduce the bandwidth and ENOB requirements of the electronics, so that simple switches or 1-Bit DACs can be used for the generation of high-bandwidth 16-QAM signals. The bandwidth improvement by three is achieved by using orthogonal signal processing with SPS in three segments ($N = 3$), and the generation of 16-QAM is enabled by an IQ configuration of the modulator in two stages (yellow and green). The simulated results for the IQ diagram of a 30 GBd, 16-QAM signal generated from 5 GHz on-off switches in such a modulator are presented in Fig.3 [18]. Another very unique advantage of orthogonal signal processing with SPS is that without any filtering, the spectrum of the generated signals has a raised cosine shape with a roll-off of $1/N$. Since the modulator has three segments, the roll-off factor is 0.33, as can be seen from the black spectrum in Fig.3. For comparison the spectrum of an on-off signal (yellow) and that of a Nyquist filtered signal (red) is shown. So, with SPS and an increasing number of segments N , Nyquist limited signals can be generated from electronic on-off inputs without any optical or electronic filtering or digital processing.

Orthogonal signal processing with SPS can be combined with an optical frequency comb to generate very high bandwidth signals, as shown for the generation of a faster than Nyquist 1.2 Tbit/s signal in a bandwidth of 662.5 GHz in Fig.4 [8]. Here 33 lines from an optical frequency comb were modulated with 22.5 GBd Nyquist QPSK data. The resulting bandwidth for each modulated comb line is with 22.5 GHz higher than the 20 GHz spacing between the lines. This can be seen by the interference between the channels in Fig.4. For the generation and reception of the QPSK data, we have used SPS with $N = 3$. Therefore, the required bandwidth of the electronics in the transmitter and receiver is $22.5 \text{ GHz}/(2 \times 3) \approx 4 \text{ GHz}$. Since these 4 GHz are much less than the 20 GHz spacing between the channels, no optical filter or special electronic post processing is required in the receiver.

Orthogonal signal processing as shown in Fig.1 can be carried out by hardware, software or electronic signal processing. Therefore, N can be quite high without increasing the complexity. In the frequency domain an SPS is a rectangular frequency comb with N equally spaced lines. This is only the case, however, if the frequency lines have the same or a linearly dependent phase. Combined with an optical frequency comb, we have used this idea for the physical layer encryption of a faster than Nyquist 270 GBd signal in a bandwidth of 252.4 GHz [9]. The frequency

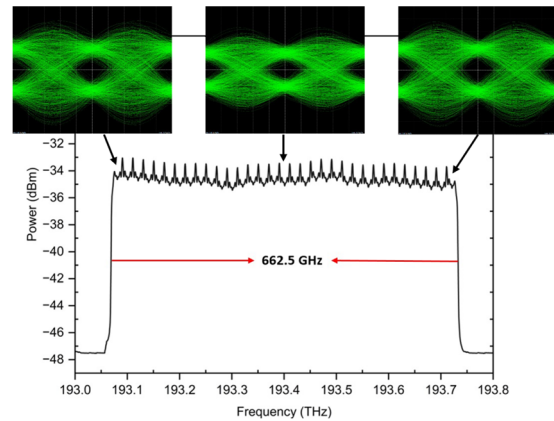


Fig.4: Faster than Nyquist transmission of 1.2 Tbit/s in a bandwidth of 662.5 GHz. The eye diagrams show the results for the left, center and right I component of the QPSK signal.

spacing between the optical comb lines was chosen in a way that neither a gap nor an interference can be seen between the modulated wavelength channels. So, the spectrum shows a smooth rectangular shape, and the wavelength channels are not visible. Each sub-signal of each wavelength channel was combined from SPS with $N = 15$. By changing the phases of the different lines in the SPS with a key, the orthogonality between the SPS was destroyed, and the superimposed signal is an arbitrary, noise-like signal with a rectangular bandwidth, which exceeds that of today's receivers. Only if the key, the bandwidth of the WDM channel and its center frequency is known, the signal can be decrypted in the receiver.

Besides bandwidth reduction, the special ad-

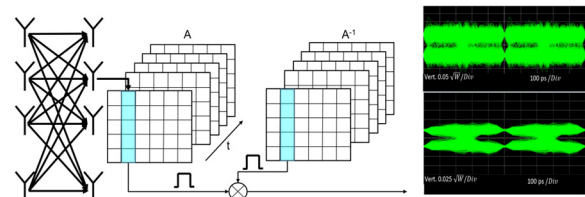


Fig.5: De-scrambling of a 6x6 MIMO signal with JPS.

vantage of JPS and DPSS is their confinement in a time frame which can be a single clock cycle. For JPS we have shown the multiplexing of six communication channels [13], and with JPS and DPSS we have presented the acceleration of optical computing [19]. As an example, Fig. 5 shows the de-scrambling for the second antenna of a 6x6 MIMO signal by parallel signal processing in a single clock cycle with JPS.

Conclusions

Here we have reviewed the generation and reception of signals with bandwidths up to the THz range with low-bandwidth electronics.

Acknowledgements

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